

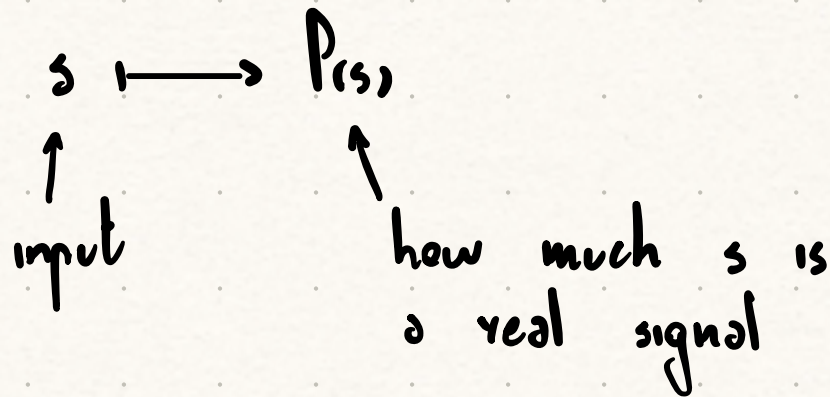
LECTURE 1

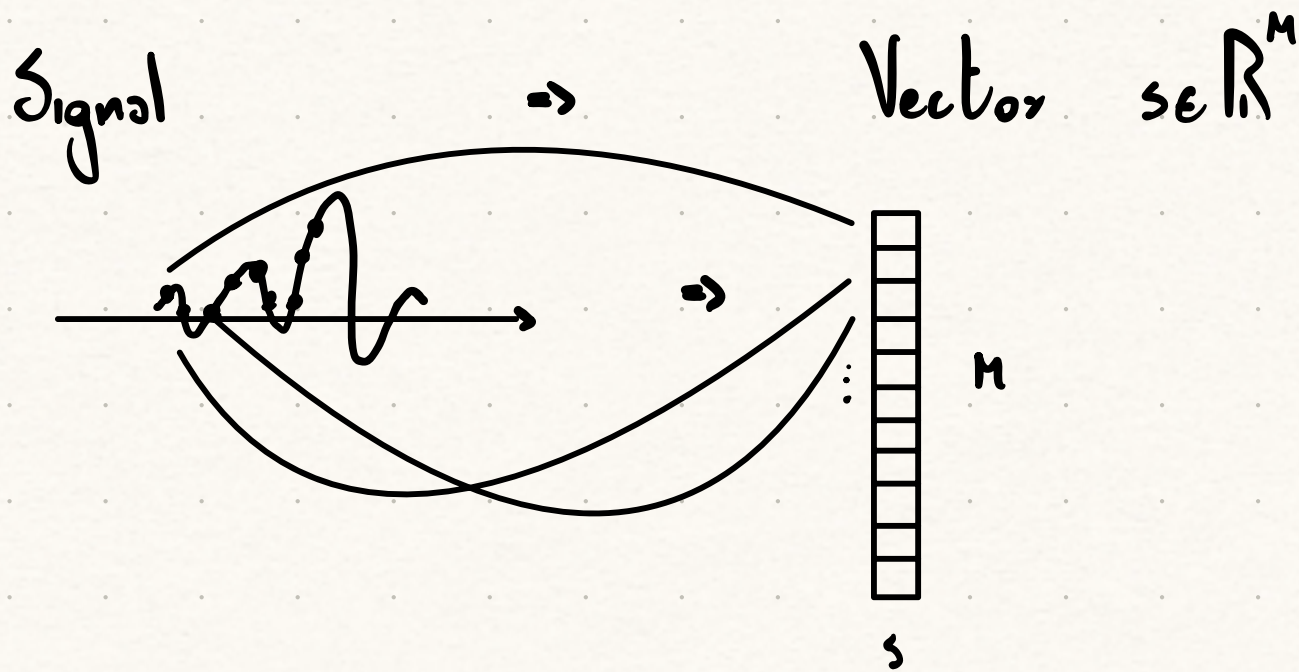
24.02.2026

Image $\rightarrow$ Vectors

We take vectors as element in a vector space that have some well defined properties

Image Priors $\hookrightarrow$ systematic way that associates s to $P(s)$

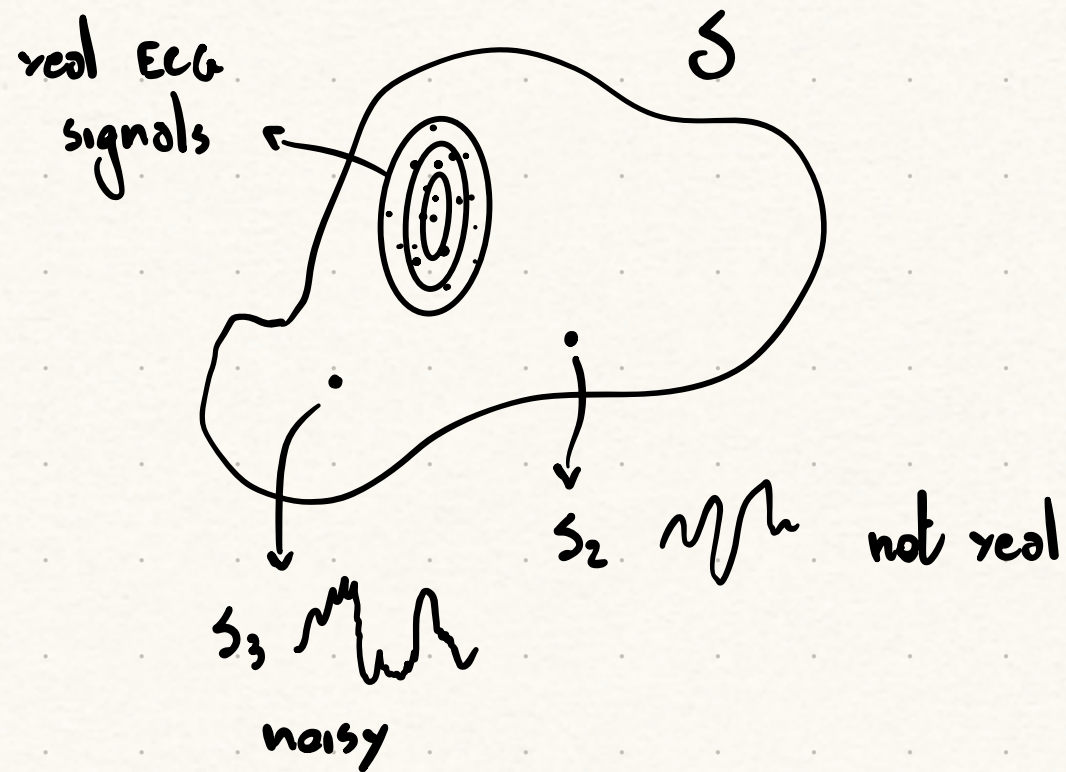




If the signal has 156 samples, s will be of 156 dim space

A prior is a function which tells me how likely a point is a signal

There is a 156-dimension SHAPE where the signal lives



S : space with dim 256

$$P(s_2) \ll P(s_1)$$

$$s \in \mathbb{R}^n \xrightarrow{\text{TRANSFORMATION}} x \in \mathbb{R}^n$$

In the transformed domain it's easier to define priors

$V = \{ \underline{v}_i \}$ Euclidean vector space
 $V \in \mathbb{R}^n$

Operations on vector spaces

sum $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ $\underline{u} + \underline{v}$

scalar mult $\mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ $\lambda \underline{v}$

inner product $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ $\langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^n \underline{u}_i \underline{v}_i$

norm $\mathbb{R}^n \rightarrow \mathbb{R}^+$ $\| \underline{u} \|_2 = \sqrt{\langle \underline{u}, \underline{u} \rangle} = \sqrt{\sum_i \underline{u}_i^2}$

Linear combinations : $\sum_i \lambda_i \underline{u}_i$ $\lambda_i \in \mathbb{R}$
 $\underline{u} \in V$

N.B. : any u of $\{u_i\}$ lives in V

Linear Independence : $\sum \lambda_i \underline{v}_i = 0 \Leftrightarrow \lambda_i = 0 \forall i$

Def: Basis of $\mathbb{R}^n$ $\{e_i\}_{i=1}^n$ set of linear independent vectors
so that

$$v = \sum x_i e_i$$

The representation x is unique if $\{e_i\}$ is an orthonormal base

$$\forall v \in \mathbb{R}^n \quad v = \sum_{i=1}^n x_i e_i \quad x \text{ is unique and } x_i = \langle v, e_i \rangle$$

Consider $\mathcal{D} \in \mathbb{R}^{n \times n}$

$$\mathcal{D} = \left[\underline{e}_1 \mid \underline{e}_2 \mid \underline{e}_3 \dots \underline{e}_n \right]$$

$$\underline{v} \mapsto \underline{x}$$

$$\underline{v} = \sum x_i \underline{e}_i$$

$$\uparrow$$
$$x_i = \langle v_i, e_i \rangle = \underline{e}_i^T \underline{v}_i$$

$$\underline{x} = \mathcal{D}^T \underline{v}$$

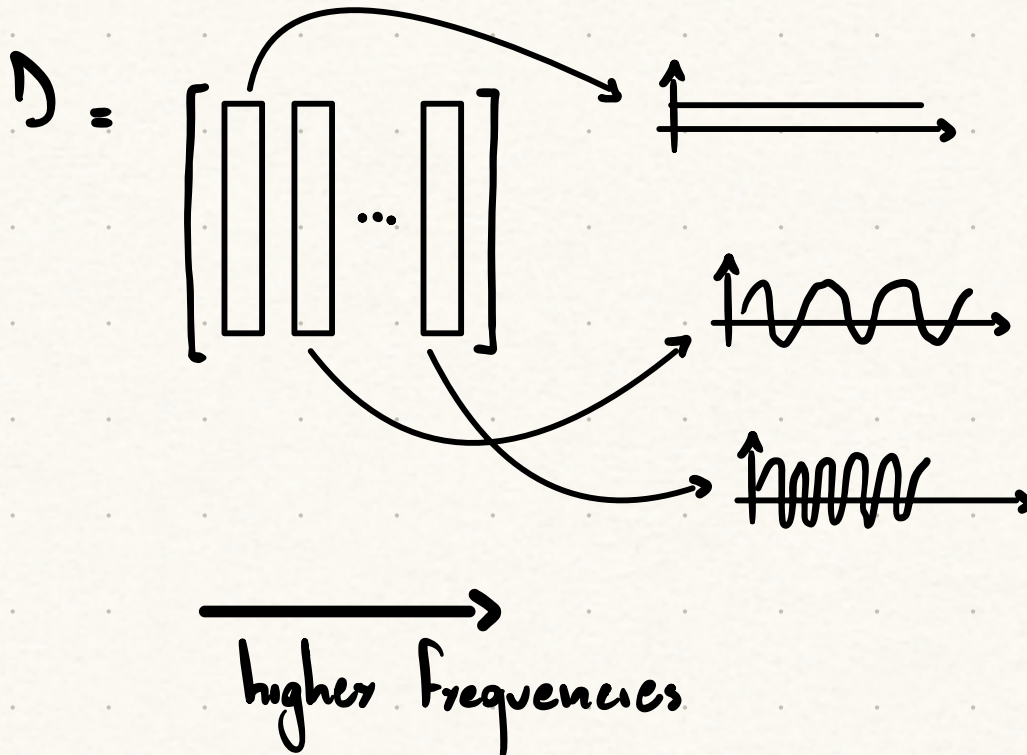
Given $\underline{x}$ you can go back to $\underline{v}$

$$\underline{s} \in \mathbb{R}^n \xrightarrow{\quad} \underline{x} = D^T \underline{s} \quad \text{TRANSF. DOMAIN}$$

D : DCT basis
↓

e_i is a sinusoidal v. spikes!

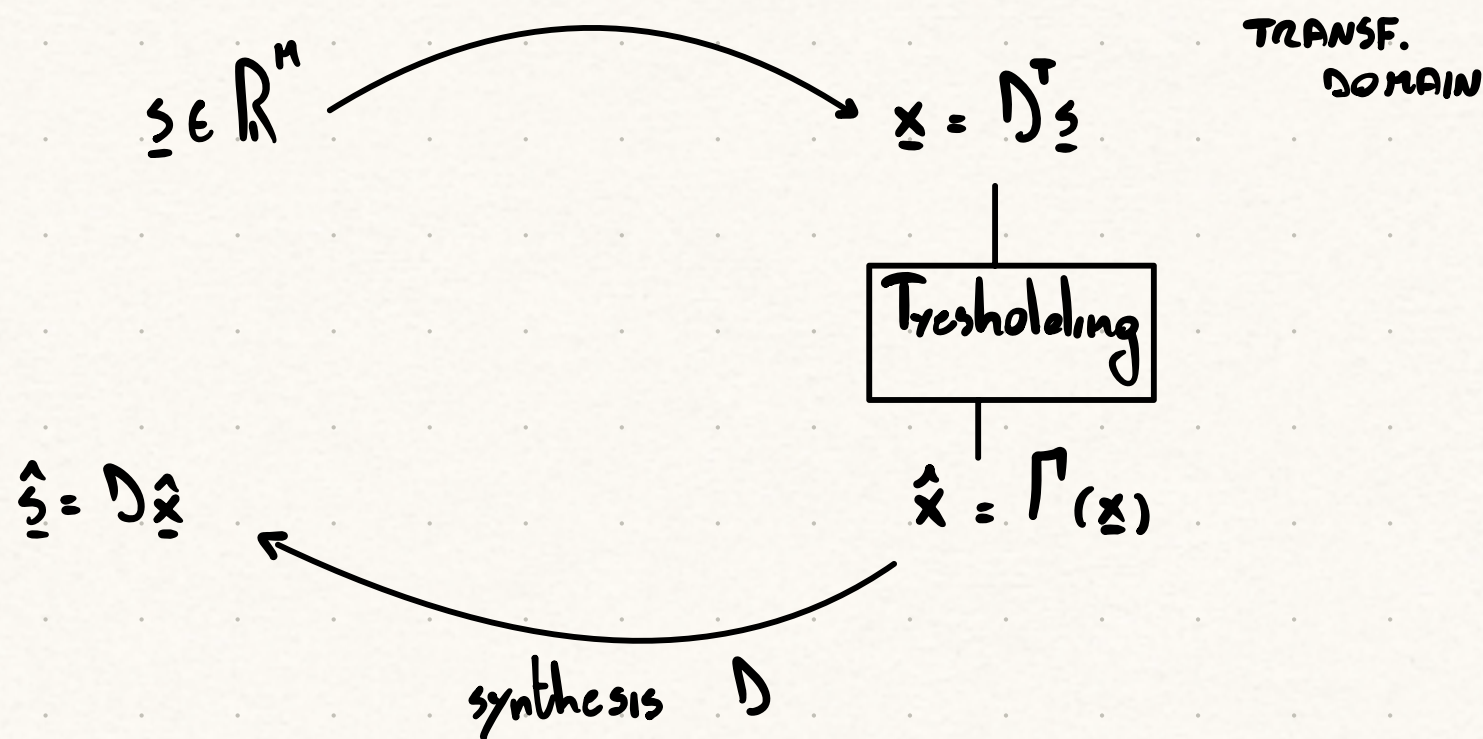
DISCRETE COSINE TRANSFORM



In a DCT basis,
signals become sparse

It is more convenient to remove noise in $\underline{x}$ domain

I can do it for example with a simple thresholding and keep like 10% of the coefficients



LECTURE 2 • DENOISING

3.03.2026

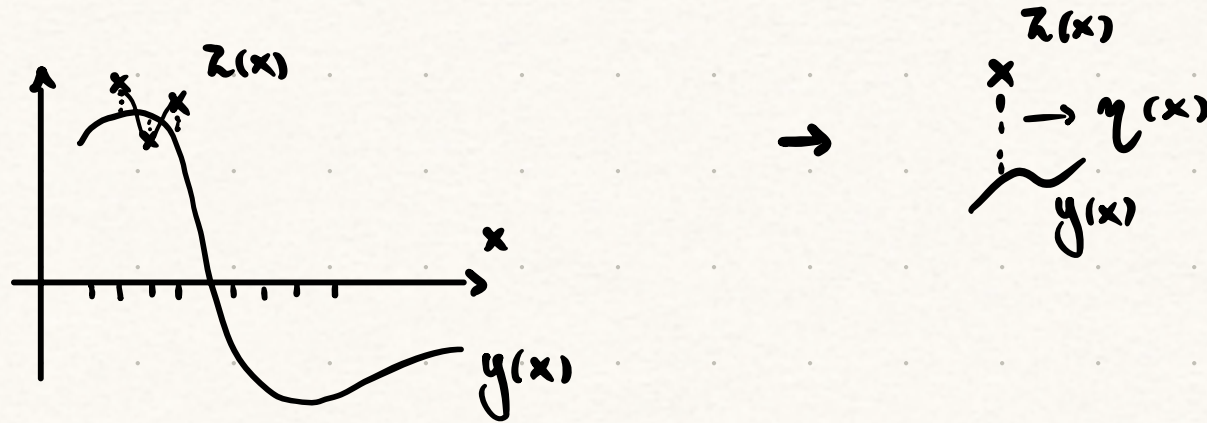
Observation model $z(x) = y(x) + \eta(x)$ $x \in X = \mathbb{N}^2$

It is a problem formulation

x is a pixel index which lives in a grid

$z(x)$ observation	}	goal is to compute $\hat{y}$ estimate of y
$y(x)$ the signal		
$\eta(x)$ noise (additive)		

Noise:



AWGN

We can assume noise as gaussian distributed signal

$$\eta(x) \sim \mathcal{N}(0, \sigma^2) \quad \text{with } \sigma^2 \text{ known}$$

We can assume η as iid realization

↓
independent identically distributed

Unrealistic assumption but good for realizing algorithms

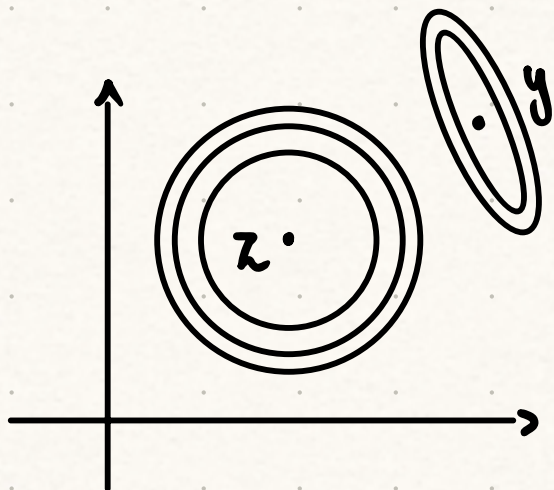
↗ cumulative density function

$$F[\eta(x_1), \eta(x_2)] = F[\eta(x_1)] F[\eta(x_2)]$$

$$E[\eta(x_1), \eta(x_2)] = E[\eta(x_1)] E[\eta(x_2)] = 0$$

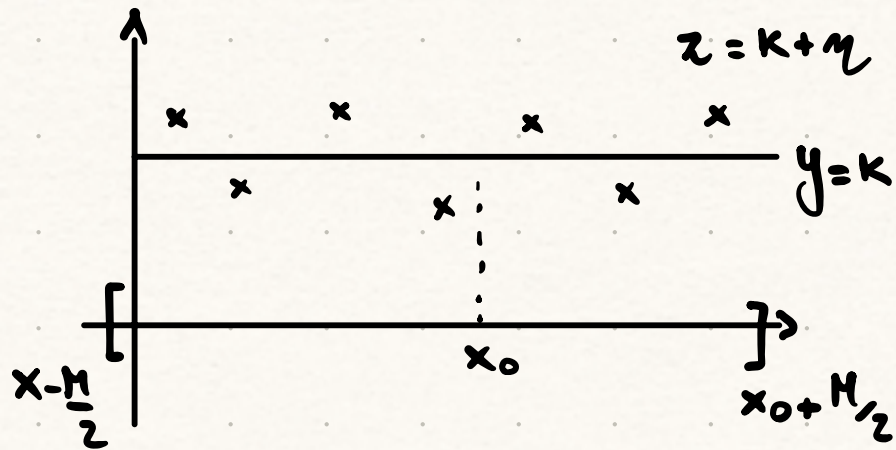
↓
Expectation

Simplest assumption for denoising, best for designing algorithm



we need a prior on y

I will assume y as locally constant



when searching for $\hat{y}(x_0)$
I will assume signal as locally constant

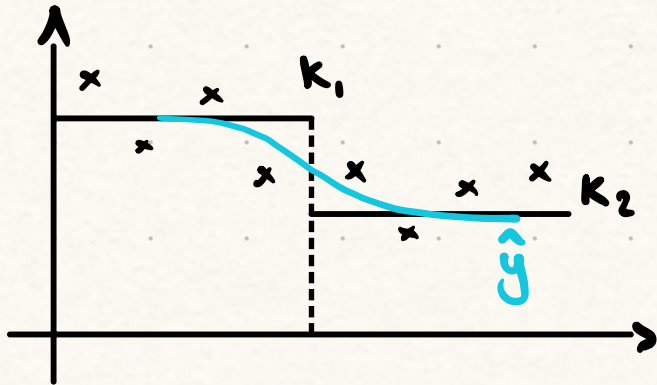
To get $\hat{y}(x_0)$ under this assumption I will average

$$\hat{y}(x_0) = \frac{1}{H} \sum_{x \in U(x_0)} z(x) = \frac{1}{H} \sum_{x \in U(x_0)} [k + \eta(x)] = k + \sum_{x \in U(x_0)} \eta(x)$$

$$S_0 \quad E[\hat{y}(x_0)] = K$$

$$\text{var} \left[\frac{1}{n} \sum_{x \in U} \eta(x) \right] = \frac{1}{n^2} \text{var} \left[\sum_{x \in U} \eta(x) \right] \stackrel{\text{iid}}{=} \frac{\sigma^2}{n}$$

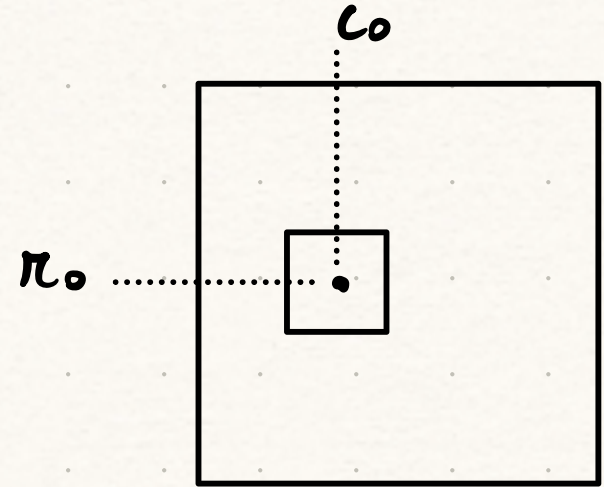
BLUE
 Best
 Linear
 Unbiased
 Estimator



$$y(x) = \begin{cases} K_1 & x \leq x_0 \\ K_2 & x > x_0 \end{cases}$$

Convolution: $\hat{y}(\underline{x}_0) = (z \otimes w)(\underline{x}_0) = \sum_{\underline{u} \in U(\underline{x}_0)} [z(\underline{x}_0 - \underline{u}) \cdot w(\underline{x}_0)]$

$$= \sum_{(u,v) \in U(\underline{x}_0)} [z(\pi_0 - u, c_0 - v) w(u, v)]$$



weights $w(u, v)$ (also called kernel or Filter)

In convolution you have the minus

You can design weights, if y is constant the best is BLUE

Convolution is a Linear and Translation Invariant operation and vice-versa.

$$\text{PSNR}(\hat{y}, y) = 10 \log_{10} \left[\frac{1}{\text{MSE}(\hat{y}, y)} \right]$$

↓

The higher the better !

It's an estimate on how good the prior is.

For the pixels at the border you do Padding.

P_s : s is sparse wrt a DCT basis

where s is a patch of the image and $D \in \mathbb{R}^{M \times M}$

DCT Denoising algorithm

inputs : • $D \in \mathbb{R}^{M \times M}$ DCT basis
• Z noisy image
• σ^2 of Z

1) $\forall s \in \mathbb{R}^M$ patch in Z compute $\underline{x} = D^T s$
which are DCT coefficients for s

2) Thresholding

$$\Gamma(\underline{x}) = \begin{cases} 0 & |x_i| < \tau \\ x_i & |x_i| > \tau \end{cases}$$

with $i = 2, \dots, M$



you start thresh.
from 2!!

at the end, $\hat{\underline{x}}$ will be a sparse vector

$$\hat{\underline{x}} = \Gamma(\underline{x})$$

3) Synthesis: $\hat{\underline{s}} = D\hat{\underline{x}}$

What is τ ? τ is typically set to $\tau = \sigma \sqrt{2 \log(M)}$

The stronger the noise the higher the threshold

In sparsity-aware weights, we multiply by $\frac{1}{\|\hat{x}\|_0}$

$\downarrow$
 L_0 norm

By using sliding process you get

- 1) Translation invariance
- 2) More priors flexibility

How to estimate sigma?

You compute $d = z * [1, -1] \Rightarrow d(x_0) = z(x_0) - z(x_{0+1})$

If the image is constant : $d(x_0) = \eta(x_0) - \eta(x_{0+1})$

$$\text{Var}(d) = 2 \text{var}(\eta) \Rightarrow \hat{\sigma}^2 = \frac{\text{var}(d)}{2}$$

↓
ESTIMATED sigma!

Another way
is with MAD : $\hat{\sigma} = \frac{\text{MAD}(z * [-1, 1])}{\sqrt{2} \cdot 0.6745}$

Median
Absolute
Deviation

LECTURE 3

Def Span of a set of vectors $\{\underline{v}_i\}_{i=1 \dots N}$ $\underline{v}_i \in \mathbb{R}^M$

$$\langle \underline{v}_1, \dots, \underline{v}_N \rangle = \{ \underline{v} \in \mathbb{R}^M, \underline{v} = \sum \lambda_i \underline{v}_i, \lambda_i \in \mathbb{R} \}$$

Recall that

Def linear independence $\sum \lambda_i \underline{v}_i = 0$ iff $\lambda_i = 0 \ \forall i$

If a space is linear independent the representation theorem holds

To check if the vectors are linearly independent we need to evaluate $\det(V)$

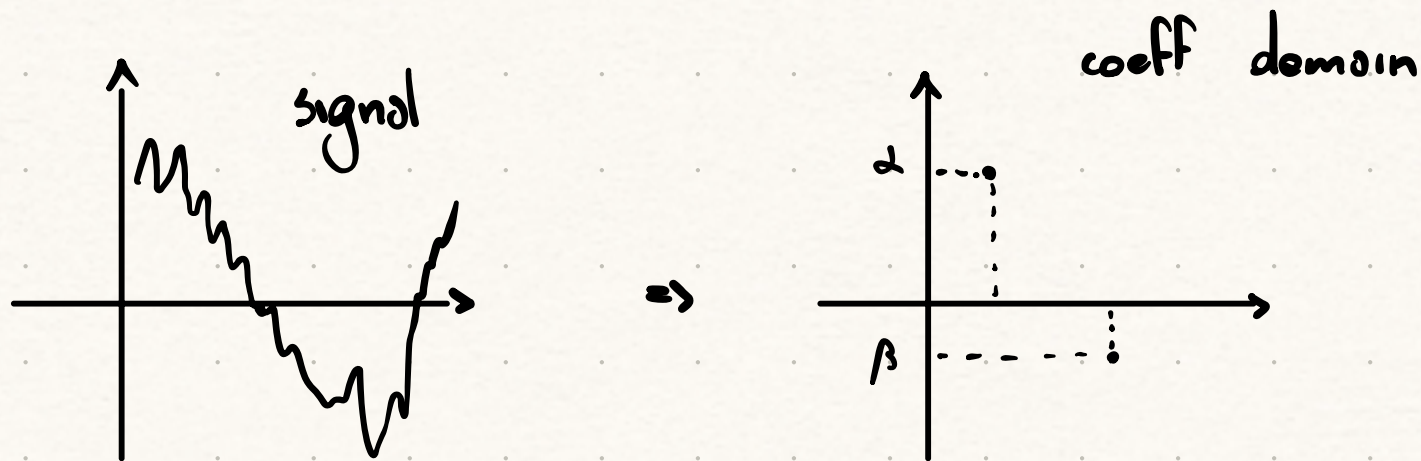
EXAMPLE where our sparsity prior doesn't work

$D_c \in \mathbb{R}^{M \times M}$ det matrix
 $C \in \mathbb{R}^{M \times M}$ canonical basis

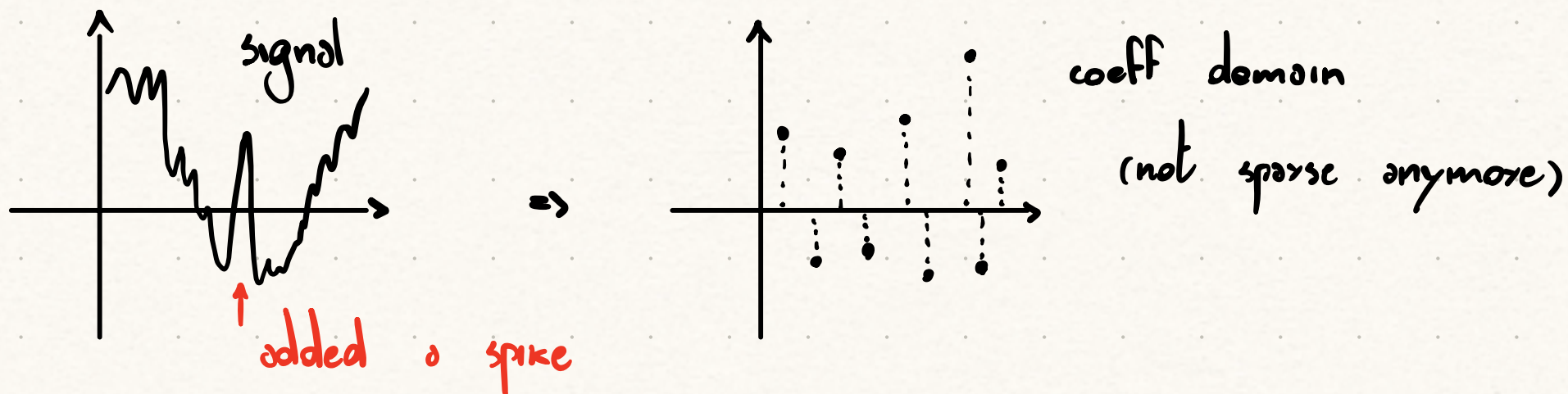
Take $\underline{z} \in \mathbb{R}^M$ s.t. $\underline{z} = \underline{z}_0 + \underline{\eta}$ where $\underline{\eta}$ iid noise

to generate a signal $\underline{z}$ sparse w.r.t D you can multiply it by a vector with few non-zero elements

$$\underline{s} = \sum_i x_i \underline{d}_i \quad \text{where} \quad x_i = 0 \text{ "often"}$$



now I add $\underline{s} = \sum_i x_i \underline{d}_i + \lambda \underline{e}_n + \underline{\eta}$



$$\begin{bmatrix} \underline{z} \end{bmatrix} = \begin{bmatrix} \overset{M}{D_c} \mid \overset{M}{C} \end{bmatrix} \begin{bmatrix} \underline{x} \end{bmatrix}_{2M}$$


 This is a
 Dictionary of size $2M$

A dictionary redundant

$$\begin{aligned} \underline{z} &\in \mathbb{R}^M \\ D &\in \mathbb{R}^{M \times 2M} \\ \underline{x} &\in \mathbb{R}^{2M} \end{aligned}$$

NOW $\underline{z}$ is sparse

Pros and Cons (+ and -)

ORTHONORMAL BASIS

$$D = D^T \quad D \in \mathbb{R}^{M \times M}$$

- + perfect reconstruction $\forall \underline{z} \in \mathbb{R}^M$
- + unique representation $\underline{x} = D^T \underline{z}$
- + closed form expression
- $\underline{z}$ might not be sparse

REDUNDANT DICTIONARY

$$D \in \mathbb{R}^{M \times N}, \quad N > M$$

- + assume perfect reconstruction
(D is full rank)
- representation $\underline{z} = D \underline{x}$ not unique
- no closed form solution
- + sparsity is a stronger prior

We will address 2 problems:

1) Sparse Coding

$$\text{Given } \underline{s} \in \mathbb{R}^M \\ \underline{D} \in \mathbb{R}^{M \times N}$$

compute $\underline{x}$ s.t. $\underline{s} = \underline{D}\underline{x}$, $\underline{x}$ sparse

2) Dictionary Learning

$$\text{Given } \underline{S} \in \mathbb{R}^M$$

compute $\underline{D} \in \mathbb{R}^{M \times N}$ which provides
sparse representation of $\underline{S}$

$$\begin{bmatrix} \vdots \\ \underline{z} \\ \vdots \end{bmatrix} = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ D \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \begin{bmatrix} w \\ \underline{x} \\ w \\ w \end{bmatrix}_N$$

N not always is $2M$
but $N > M$

$$\underline{z} = D \underline{x} \quad \begin{array}{l} \underline{z} \in \mathbb{R}^M, \quad D \in \mathbb{R}^{M \times N} \\ \underline{x} \in \mathbb{R}^N \end{array}$$

to address the first problem
we consider these as given
and this to be completed

$\underline{z} = D \underline{x}$ is like writing and solving a linear system

Theorem Rouché - Capelli

Let $D \in \mathbb{R}^{M \times N}$, $\underline{x} \in \mathbb{R}^N$, $\underline{z} \in \mathbb{R}^M$

The system $\underline{z} = D\underline{x}$ admits solution iff $\text{rk}(D) = \text{rk}([D|\underline{z}])$

It is the same as saying that by adding $\underline{z}$ you're not adding linearly independent vectors.

$$\underline{z} \in \text{span}(D)$$

The solution form an affine subspace of dimension $N - \text{rk}(D)$

$N = \text{rk}(D) \Rightarrow$ unique solution

$N > \text{rk}(D) \Rightarrow$ solution not unique, ∞ many solutions

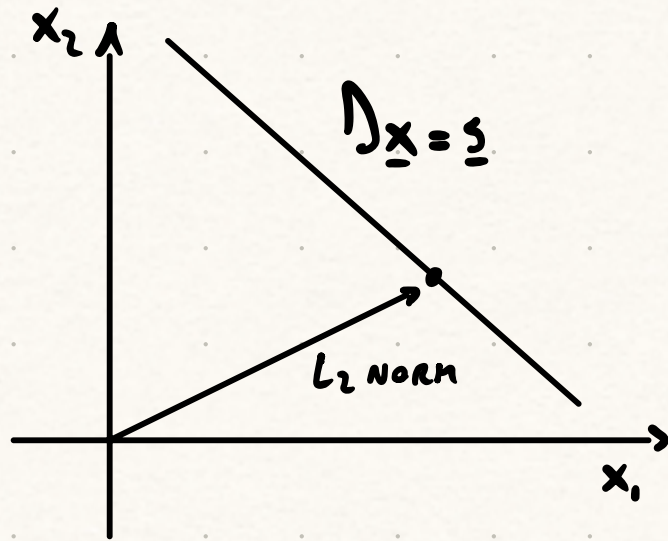
The problem is unconstrained so ... I add constraints!

Regularization

$J: \mathbb{R}^N \longrightarrow \mathbb{R}^+$ such that $J(\underline{x})$ tells "how unhappy I am of $\underline{x}$ as a solution"

I will pick $\hat{\underline{x}} = \underset{\underline{x} \in \mathbb{R}^N}{\text{argmin}} J(\underline{x})$ s.t. $\underline{y} = D\underline{x}$

$$J(\underline{x}) = \|\underline{x}\|_2^2, \quad \underline{z} = D\underline{x}$$

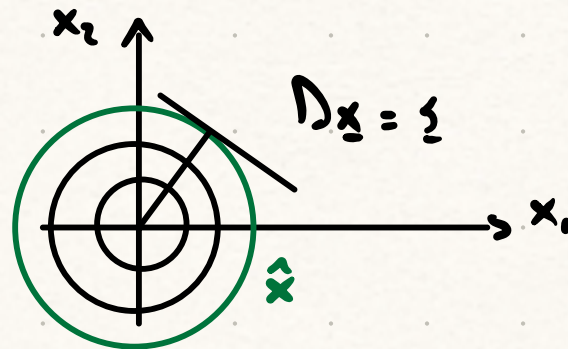


$$\underline{z} = [d_1 \ d_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= d_1 x_1 + d_2 x_2$$

it's a line!

The L_2 norm of a solution is the distance between 0 and the sol



$$\hat{\underline{x}} = \underset{\underline{x} \in \mathbb{R}^n}{\operatorname{argmin}} \left(\frac{1}{2} \| A \underline{x} - \underline{b} \|_2^2 + \frac{\lambda}{2} \| \underline{x} \|_2^2 \right)$$

Ridge regression
 $\downarrow$
 convex problem

$$\frac{\partial}{\partial \underline{x}} \left(\frac{1}{2} \| A \underline{x} - \underline{b} \|_2^2 + \frac{\lambda}{2} \| \underline{x} \|_2^2 \right) \quad \underline{x} \in \mathbb{R}^n$$

↳ it is a derivative wrt a vector

$$\frac{\partial y}{\partial \underline{x}} = \begin{bmatrix} \frac{\partial y_i}{\partial x_i} \\ \vdots \\ \frac{\partial y}{\partial x_n} \end{bmatrix}, \quad \frac{\partial y}{\partial \underline{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \dots & \frac{\partial y_n}{\partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_1}{\partial x_n} & \dots & \frac{\partial y_n}{\partial x_n} \end{bmatrix}$$

$$\frac{\partial \| \underline{x} \|_2^2}{\partial \underline{x}} = 2 \underline{x} \quad \frac{\partial \| A \underline{x} - \underline{b} \|_2^2}{\partial \underline{x}} = 2 A^T (A \underline{x} - \underline{b})$$

$$\frac{\partial}{\partial \underline{x}} \left(\frac{1}{2} \| D \underline{x} - \underline{z} \|_2^2 + \frac{\lambda}{2} \| \underline{x} \|_2^2 \right) = 0$$

$$D^T (D \underline{x} - \underline{z}) + \lambda \underline{x} = 0$$

$$D^T D \underline{x} + \mathbb{I}_N \lambda \underline{x} = D^T \underline{z}$$

$$\underbrace{(D^T D + \mathbb{I}_N \lambda)}_{\text{Invertible matrix of dimension } MN \times MN} \underline{x} = D^T \underline{z}$$

$$\hat{\underline{x}} = (D^T D + \mathbb{I}_N \lambda)^{-1} D^T \underline{z}$$

LECTURE 4 • 17 Marzo 2026

Recap

$D^T D$ is not invertible

$$\text{rk}(D^T D) < \min [\text{rk}(D), \text{rk}(D^T)]$$

$D^T D = A$ is semi-definite positive and symmetric so A can be diagonalized

$$x^T D^T D x \geq 0 \quad \forall x \quad \Rightarrow \quad y^T y \geq 0$$

true because $y^T y = \|y\|_2^2 \geq 0$

$$\hat{x} = (\underbrace{D^T D + \lambda I}_\downarrow)^{-1} D^T z$$

$$= (U^T \Lambda U + \lambda I_n) \quad \lambda_i \geq 0$$

$$= (U^T \Lambda U + \lambda U^T U)$$

$$= U^T (\underbrace{\Lambda + \lambda I}_\rightarrow) U$$

invertible diagonal matrix

NORM

$$\|x\| \quad x \in \mathbb{R}^n$$

• Positive Definite $\|x\| \geq 0 \quad \|x\| = 0 \Leftrightarrow x = 0$

• Homogeneity $\|\lambda x\| = |\lambda| \|x\| \quad \forall \lambda \in \mathbb{R}$

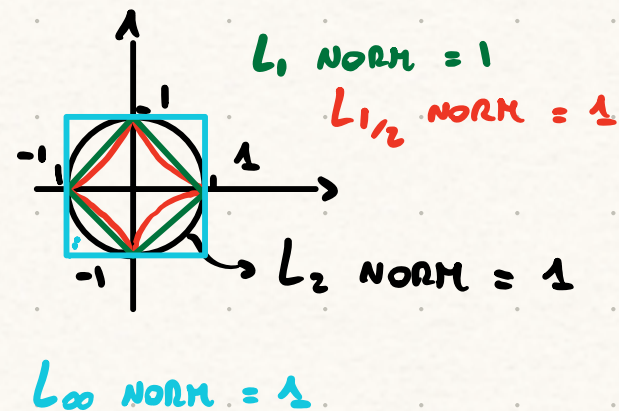
• Triangular Inequality $\|x + y\| \leq \|x\| + \|y\| \quad \forall x, y \in \mathbb{R}^n$

• Cauchy Schwarz $\langle x, y \rangle \leq \|x\|_2 \|y\|_2$

L_p norm: $\|x\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad \forall p, \quad \forall x \in \mathbb{R}^n$

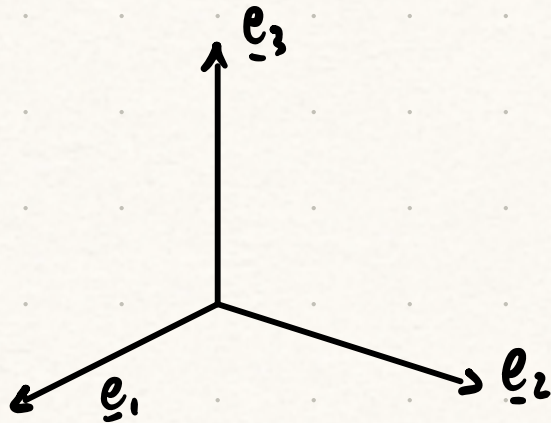
L_1 norm $\|\underline{x}\|_1 = \sum_{i=1}^N |x_i|$

L_0 norm : $\|\underline{x}\|_0 = \# \{x_i \mid x_i \neq 0\}$
 $\downarrow$
 number of



Our optimization problem becomes

$(P_0^c) \quad \hat{\underline{x}} = \underset{\underline{x} \in \mathbb{R}^N}{\operatorname{argmin}} \|\underline{x}\|_0 \quad \text{s.t.} \quad D\underline{x} = \underline{y}$



$$n=3$$

$$\underline{x} = \sum x_i e_i$$

$$\|\underline{x}_0\| = 2$$

$$D = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}_3^4$$

$$S = [d_1 | d_2 | d_3 | d_4] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

if $\|\underline{x}\|_0 = 1$ the solution lives
on one of the d_i lines

if $\|\underline{x}\|_0 = 2$ we look for the planes

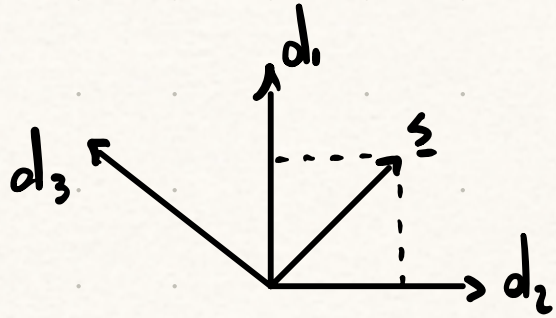
if $\|x\|_0 = 1$ $Dx = z$ I'm checking whether column of D are parallel to z $\approx N$ attempts

$\|x\|_0 = 2$ $\binom{N}{2}$ combinations $D_2 x = z$

(P_0^c) is called sparse coding problem

(P_0^c) : $\underset{x \in \mathbb{R}^N}{\operatorname{argmin}} \|x\|_0$ s.t. $Dx = z$
 $z \in \mathbb{R}^M, D \in \mathbb{R}^{M \times N}$ given

Iterative: look for the best atom to add to your solution



$$\forall j = 1, \dots, N$$

$$\begin{aligned} z_j^* &= \underset{z}{\operatorname{argmin}} \|z - z d_j\|_2^2 \\ &= \frac{\langle z, d_j \rangle}{\|d_j\|^2} = \frac{z^T d_j}{\|d_j\|^2} \quad (*) \end{aligned}$$

(*) is the min distance to an atom d_j

$$\varepsilon_j = \|z\|_2^2 - \frac{z^T d_j}{\|d_j\|^2}$$

MATCHING PURSUIT

1) INPUTS $D, \underline{z}$

INIT. $\underline{x}^{(0)} = \underline{0}$, $\underline{\pi}^{(0)} = \underline{z}$, $\omega = \{\emptyset\}$

2) SWEEP STAGE

$$\forall j = 1, \dots, N \quad \text{compute} \quad \epsilon(j) = \|\pi^{(i)}\|_2^2 - \frac{\langle \pi^{(i)}, d_j \rangle^2}{\|d_j\|_2^2}$$

$$j^* = \underset{j}{\operatorname{argmin}} \epsilon(j)$$

3) UPDATE ACTIVE SET

$$\omega^i = \omega^{i-1} \cup \{j^*\}$$

4) UPDATE THE SOLUTION

$$\underline{x}^{i+1} = \underline{x}^i + \lambda_{j^*} \underline{e}_{j^*}$$

5) UPDATE THE RESIDUAL

6) STOPPING RULE

LECTURE 5 • 24 Marzo 2026

Today we deal with dictionary learning problem

$$\begin{array}{c} \xrightarrow{N} \\ \begin{array}{c} \downarrow M \\ \left[\begin{array}{c} D \end{array} \right] \\ \downarrow \end{array} \end{array} \left[\begin{array}{c} x \end{array} \right] \downarrow N$$

$D \in \mathbb{R}^{M \times N}$ $N > M$

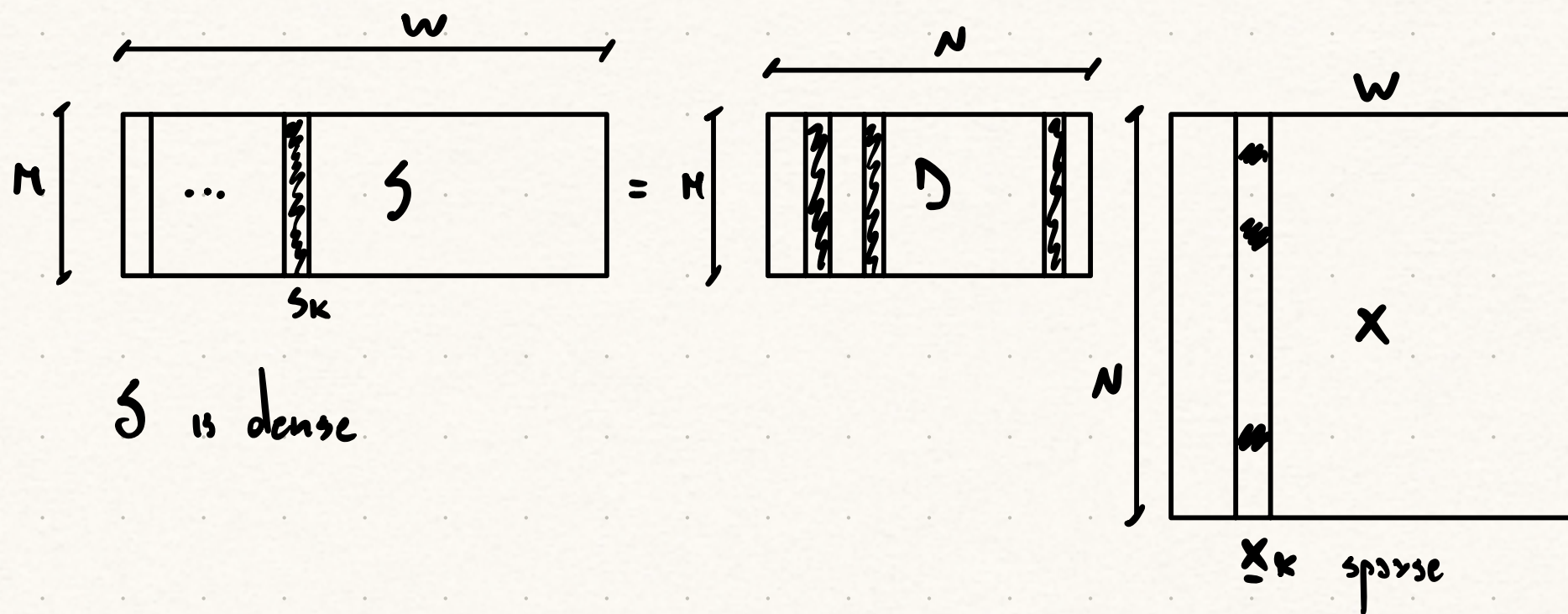
↑ to summarize:

Sparse Coding Problem given $D \in \mathbb{R}^{M \times N}$, $z \in \mathbb{R}^N$

$$(P_0)^c \quad \hat{x} = \arg\min \|x\|_0 \quad \text{s.t.} \quad \|Dx - z\|_2 < \epsilon$$

We see the image as a set of patches put columnwise

$$S \in \mathbb{R}^{H \times W}$$



$$S \in \mathbb{R}^{H \times W}, \text{ learn } D \in \mathbb{R}^{H \times N}, X \in \mathbb{R}^{N \times W} \text{ s.t. } \|X_k\|_0 < L \quad \forall k = 1, \dots, W$$

where $S = DX$

This is a matrix factorization problem

$$(DL_0): \text{Find } \hat{D}, \hat{X} = \underset{D, X}{\operatorname{argmin}} \|DX - S\|_F^2$$

F is the Frobenius norm, which is L_2 norm of matrices seen as vectors

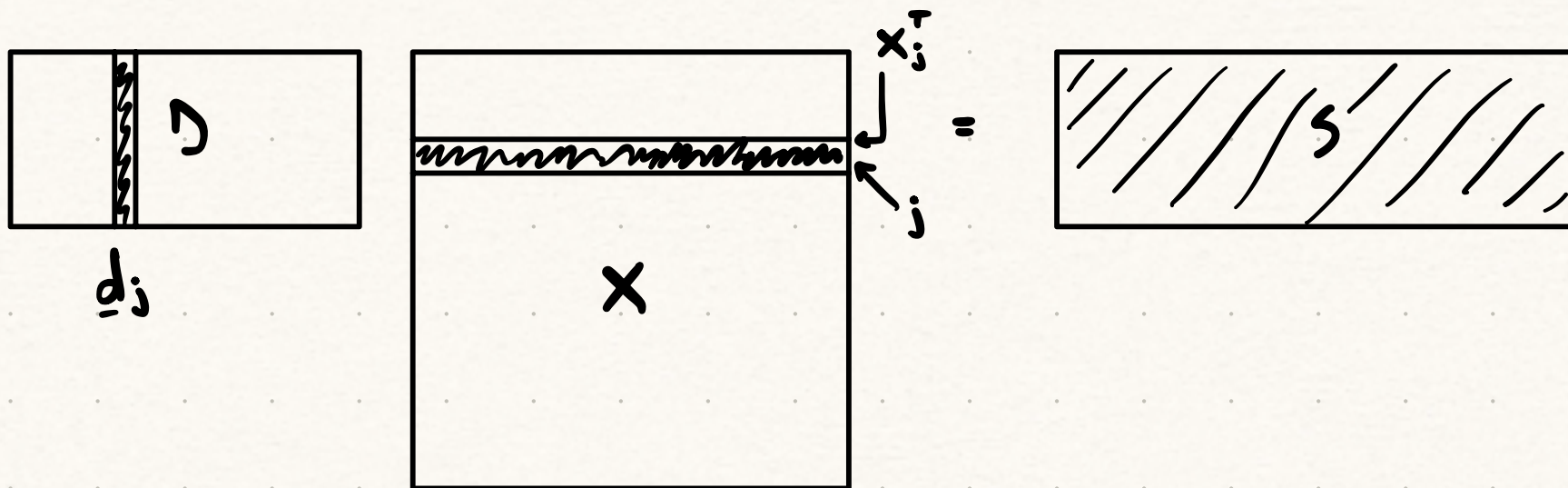
$$\|A\|_F^2 = \sum_{i,j} a_{ij}^2$$

constraints are $\|X_k\|_0 \leq L$, $\|d_j\|_2 = 1 \quad \forall k=1, \dots, W \quad \forall j=1, \dots, N$

There is no unique solution easy to find, I find something I'm happy with and I keep it 😊

K-SVD Dictionary Learning Algorithm

- 1) SPARSE CODING, UPDATE $\hat{X}$ GIVEN D FIXED
- 2) JOINT OPTIMIZATION OF D AND X
- ONE COLUMN OF D AT A TIME



$$DX = \sum_{j=1}^N \underline{d}_j \cdot \underline{x}_j^T \quad \text{i.e.} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha & \beta & \gamma \\ \delta & w & \eta \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix} \begin{bmatrix} \alpha & \beta & \gamma \end{bmatrix} + \begin{bmatrix} b \\ d \end{bmatrix} \begin{bmatrix} \delta & w & \eta \end{bmatrix}$$

Sweep columns of D

$$j_0 = 1, \dots, N \quad \argmin \| S - \sum_{j \neq j_0} (\underline{d}_j \underline{x}_j^T - \underline{d}_{j_0} \underline{x}_{j_0}^T) \|_F^2$$

The blue term is constant when I'm optimizing for j_0

$$E_{j_0} = S - \sum_{j \neq j_0} \underline{d}_j \underline{x}_j^T$$

This is the approx of S I get without $\underline{d}_{j_0}$

We get $\operatorname{argmin} \|E_{j_0} - d_{j_0} x_{j_0}^T\|_F^2$

It is like $\|A - B\|_F^2$ so it's approximating A with B

B has a peculiarity because B has rank 1!

We are looking for the best rank 1 approximation of E_{j_0}

This is done with SVD

$$\begin{array}{c} N \\ \boxed{D} \\ M \end{array} = \begin{array}{c} M \\ \boxed{U} \\ UU^T = I_M \end{array} \begin{array}{c} N \\ \boxed{\Sigma} \\ \begin{array}{l} \sigma_{ij} = 0 \quad i \neq j \\ \sigma_{ii} \geq 0 \end{array} \\ N \end{array} \begin{array}{c} N \\ \boxed{V^T} \\ VV^T = I_N \end{array}$$

$$[U, \Sigma, V] = \text{SVD}(D)$$

Best rank r approx of a matrix is the truncated SVD

$$\begin{array}{l} \arg \min \|A - B\|_F^2 \\ B, \text{rank}(B) \leq r \end{array}$$

$$B = U \Sigma_r V^T$$

Truncated SVD, $\sigma_i = 0 \quad \forall i > r$

Since we want $r_k(\lambda)$ approximation only σ_1 is non-zero
and the first column of U and row of V^T

K-SVD Summary:

INPUT: $S \in \mathbb{R}^{M \times W}$, L, N

INIT: $D = \text{rand}(M, N)$, $K=0$

NORMALIZE: $d_i = \frac{d_i}{\|d_i\|_2} \quad \forall i=1, \dots, N$

REPEAT UNTIL CONVERGENCE:

or STOPPING RULE

• $K++$

• SPARSE CODING

$$X^{(K)} = \text{OMP}(S, D^{K-1}, L)$$



- $K > K_{\max}$
- $\|S - D^{(K)} X^{(K)}\|_F^2 < T$

$$X^{(K)} = \underset{X}{\operatorname{argmin}} \|S - D^{(K-1)} X\|_F^2$$

s.t. $\|X\|_0 < L$

• DICTIONARY UPDATE

sweep columns $j_0 = 1, \dots, N$

- define support

$\Omega_0 = \text{support of } \underline{X}_{j_0}^T$

- compute residual

$$E_{j_0} = S - \sum_{j \neq j_0} \underline{d}_j - \underline{x}_j^T$$

- define P_{Ω_0} operator

- define error "reduced" to Ω_0

$$E_{j_0}^R = E_{j_0} P_{\Omega_0}$$

- compute $\mu_K(1)$ svd

- update

$$\underline{d}_{j_0}^{(K)} = \underline{\mu}_1, \quad \underline{x}_{j_0}^{(K)} = \sigma_1 \underline{v}_1$$

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SPARSE CODING:

$$(P_0)^c \quad \underset{\underline{x}}{\operatorname{argmin}} \|\underline{x}\|_0 \quad \text{s.t.} \quad \underline{D}\underline{x} = \underline{z} \\ \underline{z} \in \mathbb{R}^M, \quad \underline{D} \in \mathbb{R}^{M \times N} \quad (\text{given})$$

$$\underline{z} \xrightarrow{\text{"MP"}} \underline{x}$$

SIGNAL

$\underline{z}$

COEFFICIENTS
(SPARSE)

$\underline{x}$

ANALYSIS

$$\underline{z} \xrightarrow{\quad} \underline{x} = \underline{D}^T \underline{z}$$

$$\underline{x} = \underline{D}^T \underline{z}$$

↓

$$\hat{\underline{x}} = \Gamma(\underline{x})$$

TRESHOLD

$$\underline{z} = \underline{D}\underline{x}$$

SYNTHESIS

Synthesis is easy ☺

Analysis is much more tough ☹

$$\hat{\underline{x}}_{\text{OMP}} = \underset{\underline{x}}{\operatorname{argmin}} \left(\frac{1}{2} \| \mathbf{D}_w \underline{x} - \underline{z} \|_2^2 \right)$$

$$\hat{\underline{x}}_{\text{OMP}} = (\mathbf{D}_w^T \mathbf{D}_w)^{-1} \mathbf{D}_w^T \underline{z}$$

$$\hat{\underline{z}} = \mathbf{D} \hat{\underline{x}}_{\text{OMP}} = \mathbf{D}_w (\mathbf{D}_w^T \mathbf{D}_w)^{-1} \mathbf{D}_w^T \underline{z}$$

$$= \mathbf{A} \underline{z}$$

↓
big $M \times M$ matrix

So... is the whole loop a linear mapping?

NO it is not because the matrix A itself is designed on $\underline{s}$ through dependencies on ω .

Ridge Regression / Tikhonov Regularization

$$\hat{x} = (D^T D + \lambda I_d)^{-1} D^T \underline{s}$$

$\underline{s} \mapsto D(D^T D + \lambda I_d)^{-1} D^T \underline{s}$ it's a linear mapping

NON-ADAPTIVE

PCA and DCT are denoising based on HT so they are ADAPTIVE and NON-LINEAR

Def: $\sigma = \text{SPARK}(D)$ $D \in \mathbb{R}^{M \times N}$ $N > M$ is the smallest number of linearly dependent columns of D

RANK is the largest number of linearly independent

$$\text{Spark}(D) \leq M+1 \quad \forall D \in \mathbb{R}^{M \times N}$$

Prop $\text{spark}(\mathcal{D})$ is related to the solution of the homogeneous system

if $\mathcal{D}\underline{x} = 0$, $\underline{x} \neq 0$ then $\|\underline{x}\|_0 \geq \text{spark}(\mathcal{D})$

$\underline{x}$ s.t. $\mathcal{D}\underline{x} = 0$ so $\sum x_i d_i = 0$

Theorem

(P₀) $\hat{\mathbf{x}} = \underset{\mathbf{x}}{\text{argmin}} \|\mathbf{x}\|_0$ s.t. $\mathcal{D}\underline{x} = \underline{z}$, $\underline{z} \neq 0$ admits a unique solution
if $\|\hat{\mathbf{x}}\|_0 < \frac{1}{2} \text{spark}(\mathcal{D})$

Proof

$\mathbf{x}$ s.t. $\mathcal{D}\underline{x} = \underline{z}$ $\|\mathbf{x}\|_0 < \frac{1}{2} \text{spark}(\mathcal{D})$

let's take y s.t. $Dy = z$ $\begin{cases} Dy = z \\ D\underline{x} = z \end{cases} \Rightarrow D\underline{x} - Dy = 0$

$$D(\underline{x} - y) = 0$$

$\underline{x} - y$ is solution of the homogeneous system

$$\|\underline{x} - y\|_0 \geq \text{spark}(D)$$

for the Δ inequality $\|\underline{x} - y\|_0 \leq \|\underline{x}\|_0 + \|y\|_0$

$$\text{so } \text{spark}(D) \leq \|\underline{x} - y\|_0 \leq \|\underline{x}\|_0 + \|y\|_0 < \frac{1}{2} \text{spark}(D) + \|y\|_0$$

$$\text{so } \|y\|_0 > \frac{1}{2} \text{spark}(D)$$

LAST $\frac{1}{2}$ LAVA LUNA